11.3 Dot Product (Scalar product / Inner product)

Operations on Vectors

Let $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$, $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ and $\mathbf{w} = \langle w_1, w_2, w_3 \rangle$ be three vectors in $\mathbb{R}^3$.

(1) Addition and subtractions of vectors: $\mathbf{u} \pm \mathbf{v} = \langle u_1 \pm v_1, u_2 \pm v_2, u_3 \pm v_3 \rangle$ is a VECTOR.

(2) Multiplication of a vector by a scalar *c*: $c\mathbf{u} = \langle cu_1, cu_2, cu_3 \rangle$ is a VECTOR.

(3) Product of two vectors

(i) **DOT Product (·) :** $\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3$ which is a SCALAR.

(ii) **CROSS Product (×) :** $\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix}$ is a VECTOR.

(4) Product of three vectors

(i) **Scalar multiplication:** $\mathbf{u}(\mathbf{v} \cdot \mathbf{w})$ is a VECTOR.

(ii) **Triple Scalar product:** $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$ is a SCALAR.

(iii) **Triple Vector product:** $\mathbf{u} \times (\mathbf{v} \times \mathbf{w})$ is a VECTOR.

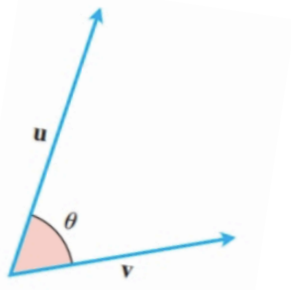
TAKE NOTE: $\mathbf{u} \cdot (\mathbf{v} \cdot \mathbf{w})$ and $\mathbf{u} \times (\mathbf{v} \cdot \mathbf{w})$ both are **meaningless**.

Definition 1: The dot product $\mathbf{u} \cdot \mathbf{v}$ (" $\mathbf{u}$ dot $\mathbf{v}$") of vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ is the scalar $\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3$.

Theorem 1: Angle between two vectors

The angle θ between two nonzero vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ is given by

$$\theta = \cos^{-1}\left(\frac{u_1v_1+u_2v_2+u_3v_3}{|\mathbf{u}||\mathbf{v}|}\right) \text{ where } 0 \le \theta \le \pi.$$



The angle between **u** and **v** given by Theorem 1 lies in the interval **[**0, π].

Remarks:

(1) If **u** and **v** both are nonzero vectors, then $\boxed{\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos\theta,\ 0 \le \theta \le \pi}$, where θ is the angle between **u** and **v**. If $\mathbf{u} = \mathbf{0}$ or $\mathbf{v} = \mathbf{0}$, then $\mathbf{u} \cdot \mathbf{v} = 0$.

(2) (i) $\mathbf{u} \cdot \mathbf{u} = \mathbf{u}^2 = |\mathbf{u}|^2$.

(ii) **u** is a unit vector $\Rightarrow \mathbf{u} \cdot \mathbf{u} = 1$.

(iii) $\mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1$.

(3) (i) $\mathbf{u} \perp \mathbf{v}$ (**u and v are orthogonal**) $\Leftrightarrow \mathbf{u} \cdot \mathbf{v} = 0$. ($\because \mathbf{u} \perp \mathbf{v} \Rightarrow \theta = \frac{\pi}{2} \Rightarrow \cos\theta = 0$)

(ii) $\mathbf{i} \cdot \mathbf{j} = \mathbf{i} \cdot \mathbf{k} = \mathbf{j} \cdot \mathbf{k} = \mathbf{j} \cdot \mathbf{i} = \mathbf{k} \cdot \mathbf{i} = \mathbf{k} \cdot \mathbf{j} = 0$.

Example 1:

(a) $< 1, -2, -1 > \cdot < -6, 2, -3 > = (1)(-6) + (-2)(2) + (-1)(-3) = -6 - 4 + 3 = -7.$

(b) $\left(\frac{1}{2}\mathbf{i} + 3\mathbf{j} + \mathbf{k}\right) \cdot (4\mathbf{i} - \mathbf{j} + 2\mathbf{k}) = \left(\frac{1}{2}\right)(4) + (3)(-1) + (1)(2) = 1.$

Proof of Theorem 1 Applying the law of cosines (Equation (8), Section 1.3) to the triangle in Figure 1, we have

$$|\mathbf{w}|^2 = |\mathbf{u}|^2 + |\mathbf{v}|^2 - 2|\mathbf{u}||\mathbf{v}| \cos\theta \qquad \text{Law of cosines}$$
$$2|\mathbf{u}||\mathbf{v}| \cos\theta = |\mathbf{u}|^2 + |\mathbf{v}|^2 - |\mathbf{w}|^2.$$

Because $\mathbf{w} = \mathbf{u} - \mathbf{v}$, the component form of $\mathbf{w}$ is $\langle u_1 - v_1, u_2 - v_2, u_3 - v_3 \rangle$. So

$$|\mathbf{u}|^2 = \left(\sqrt{u_1^2 + u_2^2 + u_3^2}\right)^2 = u_1^2 + u_2^2 + u_3^2$$
$$|\mathbf{v}|^2 = \left(\sqrt{v_1^2 + v_2^2 + v_3^2}\right)^2 = v_1^2 + v_2^2 + v_3^2$$
$$|\mathbf{w}|^2 = \left(\sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2}\right)^2$$
$$= (u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2$$
$$= u_1^2 - 2u_1v_1 + v_1^2 + u_2^2 - 2u_2v_2 + v_2^2 + u_3^2 - 2u_3v_3 + v_3^2$$

and

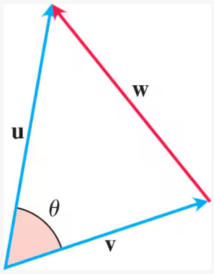
$$|\mathbf{u}|^2 + |\mathbf{v}|^2 - |\mathbf{w}|^2 = 2(u_1v_1 + u_2v_2 + u_3v_3).$$

Therefore,

$$2|\mathbf{u}||\mathbf{v}| \cos\theta = |\mathbf{u}|^2 + |\mathbf{v}|^2 - |\mathbf{w}|^2 = 2(u_1v_1 + u_2v_2 + u_3v_3)$$
$$|\mathbf{u}||\mathbf{v}| \cos\theta = u_1v_1 + u_2v_2 + u_3v_3$$
$$\cos\theta = \frac{u_1v_1 + u_2v_2 + u_3v_3}{|\mathbf{u}||\mathbf{v}|}.$$

Since $0 \le \theta < \pi$, we have $\theta = \cos^{-1}\left(\frac{u_1v_1 + u_2v_2 + u_3v_3}{|\mathbf{u}||\mathbf{v}|}\right)$. ■

Figure 1



The parallelogram law of addition of vectors gives $\mathbf{w} = \mathbf{u} - \mathbf{v}$.

Example 2:

Find the angle between the vectors $\mathbf{u} = \sqrt{3}\mathbf{i} - 5\mathbf{j}$ and $\mathbf{v} = \sqrt{3}\mathbf{i} + \mathbf{j} - 2\mathbf{k}$.

Solution:

The angle between the vectors $\mathbf{u}$ and $\mathbf{v} = \cos^{-1}\left(\frac{(\sqrt{3})(\sqrt{3})-5(1)+0}{\sqrt{3+25}\sqrt{3+1+4}}\right) = \cos^{-1}\left(\frac{-1}{\sqrt{56}}\right) \approx 1.70$ rad.

Perpendicular (Orthogonal) Vectors

Definition 2: Vectors **u** and **v** are orthogonal if $\mathbf{u} \cdot \mathbf{v} = 0$.

Example 3:

(a) $\mathbf{u} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $\mathbf{v} = 2\mathbf{j} + 4\mathbf{k}$ are orthogonal because

$$\mathbf{u} \cdot \mathbf{v} = (3\mathbf{i} - 2\mathbf{j} + \mathbf{k}) \cdot (2\mathbf{j} + 4\mathbf{k}) = (3)(0) + (-2)(2) + (1)(4) = 0.$$

(b) **0** is orthogonal to every vector **u** since because

$$\mathbf{0} \cdot \mathbf{u} =< 0, 0, 0 >\cdot< u_1, u_2, v_3 >= (0)(u_1) + (0)(u_2) + (0)(u_3) = 0.$$

Example 4:

Let $P(1, 1, -1)$, $Q(0, 3, -2)$, $R(-2, 1, 2)$ and $S(k, 0, 2)$ be four points in space. Find the value of k if

(i) $\overrightarrow{PQ}$ is orthogonal to $\overrightarrow{PS}$

(ii) $\overrightarrow{RS}$ is a unit vector.

Dot Product Properties

Properties of the Dot Product

If **u**, **v**, and **w** are any vectors and c is a scalar, then

1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
2. $(c\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (c\mathbf{v}) = c(\mathbf{u} \cdot \mathbf{v})$
3. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$
4. $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$
5. $\mathbf{0} \cdot \mathbf{u} = 0.$

Vectors Projections

The **vector projection** of $\mathbf{u} = \overrightarrow{PQ}$ onto a nonzero vector $\mathbf{u} = \overrightarrow{PS}$ (Figure 1) is the vector $\overrightarrow{PR}$ determined by dropping a perpendicular from Q to the line PS. The notation for this vector is

$\text{proj}_{\mathbf{v}}\ \mathbf{u}$ ("the vector projection of **u** onto **v**").

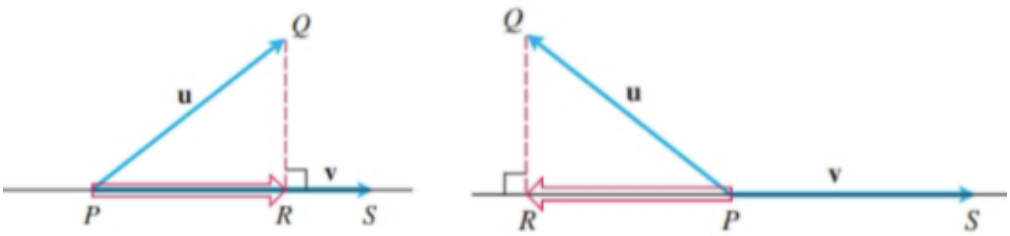
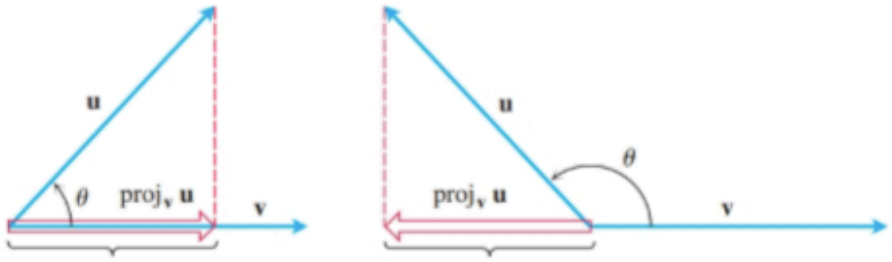


Figure 1 The vector projection of **u** onto **v.**

If **u** represents a force, then $\text{proj}_{\mathbf{v}}\ \mathbf{u}$ represents the effective force in the direction of **v.**
If the angle θ between **u** and *v* is acute, $\text{proj}_{\mathbf{v}}\ \mathbf{u}$ has length $|\mathbf{u}| \cos\theta$ and direction $\mathbf{v}/|\mathbf{v}|$.
If the angle θ between **u** and *v* is obtuse, $\cos\theta < 0$ and $\text{proj}_{\mathbf{v}}\ \mathbf{u}$ has length $-|\mathbf{u}| \cos\theta$ and direction $-\mathbf{v}/|\mathbf{v}|$. In both case,

$$\text{Proj}_{\mathbf{v}}\ \mathbf{u} = (|\mathbf{u}|\cos\theta)\frac{\mathbf{v}}{|\mathbf{v}|} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|}\right)\frac{\mathbf{v}}{|\mathbf{v}|} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2}\right)\mathbf{v}$$



(a) Length = $|\mathbf{u}|\cos\theta$ (b) Length = $-|\mathbf{u}|\cos\theta$

Figure 2 The length of $\text{proj}_{\mathbf{v}}\,\mathbf{u}$ is (a) $|\mathbf{u}|\cos\theta$ if $\cos\theta \geq 0$ and (b) $-|\mathbf{u}|\cos\theta$ if $\cos\theta < 0$.

The number $|\mathbf{u}|\cos\theta$ is called the **scalar component of u in the direction of v**.

To summarize,

The vector projection of **u** onto **v** is the vector $\text{Proj}_{\mathbf{v}}\,\mathbf{u} = \left(\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|}\right)\frac{\mathbf{v}}{|\mathbf{v}|} = \left(\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|^2}\right)\mathbf{v}$.

The scalar component of **u** in the direction of **v** is the scalar $|\mathbf{u}|\cos\theta = \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|} = \frac{\mathbf{u}}{|\mathbf{v}|}\cdot\mathbf{v}$

Example 5: Find the vector projection of $\mathbf{u} = 6\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$ onto $\mathbf{v} = \mathbf{i} - 2\mathbf{j} - 2\mathbf{k}$ and the scalar component of **u** in the direction of **v.**

Solution:

The vector projection of **u** onto **v** is the vector

$$\begin{aligned}\text{proj}_{\mathbf{v}}\mathbf{u} &= \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|^2}\mathbf{v} = \frac{\mathbf{u}\cdot\mathbf{v}}{\mathbf{v}\cdot\mathbf{v}}\mathbf{v} = \frac{6-6-4}{1+4+4}(\mathbf{i}-2\mathbf{j}-2\mathbf{k}) \\ &= -\tfrac{4}{9}(\mathbf{i}-2\mathbf{j}-2\mathbf{k}) = -\tfrac{4}{9}\mathbf{i} + \tfrac{8}{9}\mathbf{j} + \tfrac{8}{9}\mathbf{k}.\end{aligned}$$

The scalar component of **u** in the direction of **v** is the scalar

$$\begin{aligned}|\mathbf{u}|\cos\theta &= \mathbf{u}\cdot\frac{\mathbf{v}}{|\mathbf{v}|} = (6\mathbf{i}+3\mathbf{j}+2\mathbf{k})\cdot\left(\tfrac{1}{3}\mathbf{i} - \tfrac{2}{3}\mathbf{j} - \tfrac{2}{3}\mathbf{k}\right) \\ &= 2 - 2 - \tfrac{4}{3} = -\tfrac{4}{3}.\end{aligned}$$

Example 6: Verify that the vector **u** - $\text{proj}_{\mathbf{v}}\,\mathbf{u}$ is orthogonal to the projection vector $\text{proj}_{\mathbf{v}}\,\mathbf{u}$.

Solution:

$$\begin{aligned}\left(\mathbf{u} - \text{proj}_{\mathbf{v}}\mathbf{u}\right)\cdot\mathbf{v} &= \mathbf{u}\cdot\mathbf{v} - \left(\frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|^2}\mathbf{v}\right)\cdot\mathbf{v} && \text{Definition of } \text{proj}_{\mathbf{v}}\mathbf{u} \\ &= \mathbf{u}\cdot\mathbf{v} - \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|^2}(\mathbf{v}\cdot\mathbf{v}) && \text{Dot product property (2)} \\ &= \mathbf{u}\cdot\mathbf{v} - \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|^2}|\mathbf{v}|^2 && \mathbf{v}\cdot\mathbf{v} = |\mathbf{v}|^2 \\ &= \mathbf{u}\cdot\mathbf{v} - \mathbf{u}\cdot\mathbf{v} = 0.\end{aligned}$$

Hence, the vector **u** - $\text{proj}_{\mathbf{v}}\,\mathbf{u}$ is orthogonal to the projection vector $\text{proj}_{\mathbf{v}}\,\mathbf{u}$.